

Area of application

This document applies to: International

☐ Mechanical	☒ Induction	☐ Installation	☐ Update	☐ Precast concrete tower-Installation
☒ Electrical	☐ Rotor blade	☐ Grid connection	☐ Concrete repair	☐ Precast concrete tower-Prestressing

☒ E-10	☒ E-26	☒ E-40/6.44 E2	☒ E-66	☒ E-82	☒ E-115
☒ E-12	☒ E-30	☒ E-40/6.44 E3	☒ E-66/18.66	☒ E-82 E2	☒ E-126
☒ E-15	☒ E-30/3.30	☒ E-44	☒ E-66/18.70	☒ E-82 E3	☒ E-126 EP4
☒ E-16	☒ E-32	☒ E-48	☒ E-66/20.70	☒ E-82 E4	
☒ E-17	☒ E-33	☒ E-53	☒ E-70 E4	☒ E-92	
☒ E-18	☒ E-40	☒ E-58/10.58	☒ E-70 E4-2	☒ E-101	
☒ E-20	☒ E-40/6.44	☒ E-58/12.58	☒ E-70 E4-3	☒ E-112	

☒ CS12	☒ CS20	☒ CS32	☒ CS44b	☒ CS58b	☒ CS70b	☒ CS112a
☒ CS16	☒ CS30	☒ CS40	☒ CS48a	☒ CS66	☒ CS82a	☒ CS126a
☒ CS17	☒ CS31a	☒ CS44a	☒ CS58a	☒ CS70a	☒ CS101a	

Document details

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Revision table

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000	2014-08-01	-	Document created
001	2015-08-18	1	Updated to reflect latest knowledge
		1.5	New material added
		3	Work step 7 revised
		4.1	Work steps 1, 6, 7 and 8 revised
		4.2.1	Work step 2 revised
		4.2.4	Added work step 4, table
		5	Updated to reflect latest knowledge
		6	Section added
001a		4.2.4	Change: kW instead of kWh
002	2017-06-19	1.2	Section revised
		1.3	New section added
		1.5	New measuring devices added
		3	Section revised
		4.1	Section revised
		4.2	Section revised

Revision	Date	Section	Change
		4.2.4	Section revised
		5	Section revised

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1 General information

Be sure to read and understand all safety instructions before commencing work.

The photos/figures in this document serve as illustrations and may differ from the actual components.

The purpose of these instructions is to help with the insulation testing of generators.

1.1 When to carry out work

The instructions provided in this document are to be carried out in the event of a malfunction or during generator inspection.

1.2 Labour input for insulation testing

The labour input for insulation testing depends on the number of generator systems. The time required for a generator with two systems can be found in the following table. The insulation measurement is primarily used for rapid troubleshooting.

Scheduled duration of operation (hours)	Number of technicians required	Total man-hours
1	2	2

1.3 Labour input for PI-measurement

Generator variant	Scheduled duration of operation (hours)	Number of technicians required	Total man-hours
All generators with 2 systems	1.5	2	3
All asymmetrical generators with 4 systems	2	2	4
E-126 and E-112 generators	3.5	2	7
All asymmetrical generators with 8 systems	4.5	2	9

1.4 Reason for instructions

The purpose of these instructions is to standardise the procedure for taking generator measurements. Insulation testing is used to determine whether a generator has a phase fault or earth fault. These instructions can also be used to determine the condition of the generator's insulation.

1.5 Required tools (only order if missing!)

Quantity		Required tools	SAP
		Standard tools	
1	or	Insulation measurement device- MEGGER® MIT420/2 (the PI measurement can be performed using this device)	168104
1		Insulation measurement device CP- MIT410-ER 2 (the PI measurement can be performed using this device)	136511
1	or	Installation tester measuring device FLUKE®1654B	114723
1		Insulation tester METRISO® 1000D	25110
1		Set of measuring leads CalPlus®	166927
1		Humidity/temperature measuring device TESTO®605-H1	166065
12		Conical spring washer DIN6796 10 x 23 x 2.5 mech.galv. (for rectifiers with M10 bolts)	27296
12		Conical spring washer DIN6796 12 x 29 x 03 mech.galv. (for rectifiers with M12 bolts)	27297
12		Conical spring washer DIN6796 16 HDG (for rectifiers with M16 bolts)	61191
8		Conical spring washer DIN6796 20 x 45 x 05 mech. galv.	27298
Only order via Technical Service Dispatch if approved			
1		Testbox high voltage 0-4 kV HIPPE KG	53873
1		Transport case f. Hippe high-voltage tester	99074

2 Safety instructions

Improper conduct in or around the WEC

Death or severe injury

- ⇒ ENERCON health and safety regulations apply.
- ⇒ Follow all relevant ENERCON instructions for safe use as well as the manufacturers' operating manuals and instructions.
- ⇒ Should the safety standards of the respective country be more stringent than those required by ENERCON, the national regulations apply.

Electric current

Death or serious injury from electric arcs and electric shocks

Follow the 5 safety rules:

- ⇒ Disconnect
- ⇒ Secure isolation
- ⇒ Prove dead
- ⇒ Earth and short-circuit
- ⇒ Cover or close off adjacent live parts

Rotor may idle if not locked

Death or severe injury

- ⇒ Do not work on or near the rotor unless the rotor lock is set.
- ⇒ Before setting the rotor lock, check whether conditions allow for the rotor lock to be set.
- ⇒ Keep a safe distance from rotating parts.
- ⇒ Do not enter the rotor area unless the rotor lock is set.
- ⇒ Before releasing the rotor lock, ensure that no one is in the rotor area.
- ⇒ Do not set the rotor lock/rotor brake if wind speeds exceed 16 m/s over a 10-minute-mean.

3 Preparation

1. Switch the WEC to maintenance mode.
2. Brake the rotor.



On wind energy converters of type E-126 there is no rotor lock. The brakes need to be used.

3. Lock the rotor.



On type E-40 WECs, EMERGENCY SWITCHING OFF needs to be pressed on the nacelle sub-distribution system.

On type E-CS31 WECs, EMERGENCY SWITCHING OFF needs to be pressed on the nacelle control cabinet.

4. Switch off main switch -QA10 (formerly -S01) on the nacelle control cabinet (= 015).
5. Prepare the test device. Check if the battery is charged.
6. Do a visual inspection of the entire generator on both the machine house side and the rotor head side.
7. Check the entire stator windings for damage, dirt or foreign objects.
8. Check all pole shoes for damage and burn marks.
9. Fix any damage that can be repaired immediately. If any damage cannot be repaired due to its severity, a damage report needs to be drawn up.

4 Measurement

4.1 Insulation measurement on the stator



Fig. 1: Rectifier with fuses



Fig. 2: Rectifier with shunt resistors



Fig. 3: Filter cabinet on generator with opened circuit breaker panel

1. Remove the fuses from the rectifiers (Fig. 1). If necessary, unscrew the shunt resistors on the rectifiers on one side (Fig. 2). If there are no fuses or shunt resistors on the rectifier, only undo the rectifier bolt connections.
2. Create a space between the plates so that measurements are only taken on the generator side.



For type E-40 WECs, the rectifiers are located in the power cabinets in the tower base.



The conical spring washers on the loose bolt connections are not used again. New conical spring washers must be used.

3. Remove the fuse-switches from the filter cabinets (Fig. 3, optional).
4. Disconnect the surge arresters individually.



DANGER!

Electric current

Death or serious injury from electric arcs and electric shocks

⇒ Keep out of the generator area during testing.

5. In the rectifier, measure any phase of system 1 against earth (PE) with a voltage of 1000 V.
6. In the rectifier, measure any phase of system 2 against earth (PE) with a voltage of 1000 V.
7. In the rectifier, measure a phase of system 1 against any phase on system 2 with a voltage of 1000 V.
 - ➔ The measured value is high-ohmic.
 - ➔ The individual measured values of both systems against earth are roughly the same.
 - ➔ The resistance between the systems is approx. twice as high as the measured values of system 1 or system 2 against earth.
8. Check if the measured value is plausible (1).

Generator condition	Measured value
Stator near new / dry / warm	MΩ range
Generator damp	KΩ range

Table 2: Plausibility of measured values

In the following scenarios, contact the Technical Service Dispatch to determine how to proceed:

- Obvious damage to stator or rotor.
- Large differences between measurement results of system 1 and system 2 against chassis ground.
- The measured resistance for system 1 against system 2 is far below the measured values of system 1 and system 2 against chassis ground.

4.2 Insulation measurement and rotor resistance measurement

In the event of faults on the rotor, take the following measurements to determine possible damage:

- Insulation resistance measurement
- Rotor resistance measurement

Measurements can be taken in the rotor sub-distribution system, at the rotor disk connection or on the excitation controller. The measurement taken on the excitation controller is also used to measure and check the slip ring unit.

4.2.1 Insulation testing on excitation controller

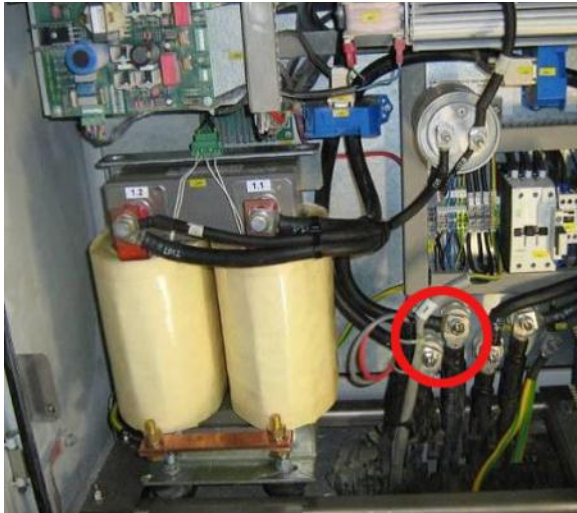


Fig. 4: Overview of excitation controller

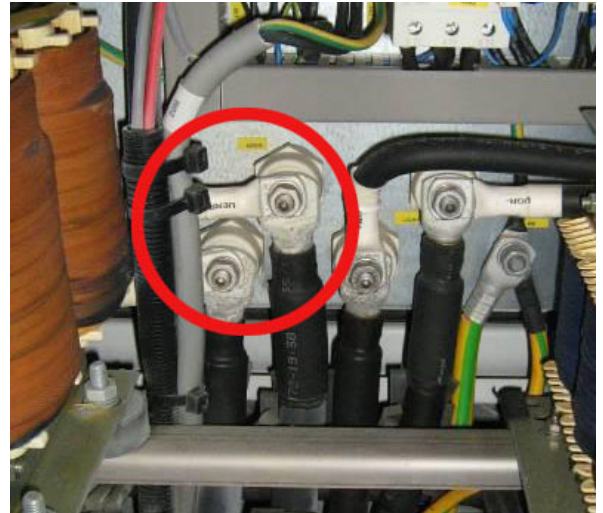


Fig. 5: Connection points on excitation controller

1. Disconnect connecting cables Uerr + and Uerr- (Fig. 5).
2. Measure Uerr + against PE using a voltage of 1000 V:
3. Evaluate the measured value as per Table:

Measured value	Result	Work step
MΩ range	OK	-
< MΩ range	Not OK	Notify Technical Service Dispatch.

Table 3: Measurement results

4.2.2 Insulation test at connection point on rotor disk (optional)



Fig. 6: Housing cover of rotor connection point

For the current product range > E-70 E4 the rotor connection points are located on the rotor disk.

1. Remove the housing cover (Fig. 6).
2. Disconnect connecting cables U_{err+} and U_{err-} as well as the cables to the surge protection device (Fig. 7).
3. Measure the rotor against earth (PE).
4. Evaluate the measured value as per Table 4:

Measured value	Result	Work step
MΩ range	OK	-
< MΩ range	Not OK	Notify Technical Service Dispatch.

Table 4: Measurement results

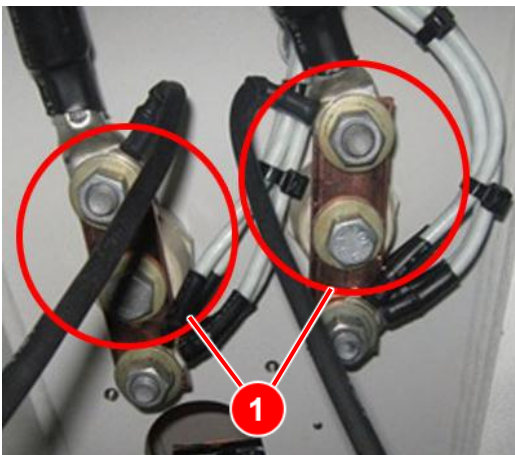


Fig. 7: Rotor connection point

1 Disconnect here

4.2.3 Insulation testing on rotor sub-distribution system



Fig. 8: Terminal block in rotor sub-distribution system

1. In the rotor sub-distribution system, disconnect the direct voltage cable on the rotor side terminal block (+ and - or blue and brown) (Fig. 8).
⇒ The measurement covers only the rotor.
2. Measure the rotor against earth (PE).
3. Evaluate the measured value as per Table 4.

Measured value	Result	Work step
MΩ range	OK	-
< MΩ range	Not OK	Notify Technical Service Dispatch.

Table 4: Measurement results

4.2.4 Measuring rotor resistance

The rotor resistance can be measured at the following points:

- On the excitation controller
 - On the rotor sub-distribution system
 - At the rotor disk connection point
1. Disconnect the rotor connections at the selected measurement point.
 2. Connect the two test leads to the positive and negative poles of the rotor connections.
 3. Measure the rotor resistance.
 4. Check if the measured value is plausible. Table 5 lists the rotor resistance values of the different generator types.

Generator version	Power in kW	≈Rotor resistance in [Ω] at 20 °C (68 °F)
G-33/3.3-G1 and G2	300	10.5
E-40/1	500	10.5
E-40/2 (depending on version)	500	8.5 - 20
E-40/6 G1	600	9.5
E-40/6 G2; and E-40/6 G2 E2 and E-40/6 G2 E3	600	9.5
G-48-G2	800	7.4
G-53/8-G1	800	5.9
G-48/9-G3 or G-44/9-G1	900	7.4
G-48/-G1	800	9.3
G-58/10-G1	1000	6.0
G-66/18-G1	1800	2.6
G-66/20-G2	2000	2.6
G-70/20-G1	2300	2.6
G-82/20-G1	2000	1.6
G-82/30-G1	3000	1.6
G-70/20-G2	2300	2.6
G-70/20-G3 and -G4	2300	2.0
G-82/30-G2	3000	1.2
G-82/23-G1	2300	1.6
G-82/30-G3 and -G4	3000	1.2
G-92/23-G1	2300	1.2
G-101/30-G1	3000	1.1
G-101/30-G2	3000	1.1
G-101/30-G3	3000	1.3
G-101/30-G4	3000	1.1
G-115/30-G1	3000	1.1
G-115/30-G2	3000	1.1
E-112	4500	1.1
E-126 EP4-GE-01	4500	1.2
G-126/75-G1	7500	1.1
G-126/75-G2	7500	1.1

Table 5: List of rotor resistance values of different generators

5 Polarisation index test (optional)

5.1 Defining and preparing the measurement

The PI-measurement can only be performed with the insulation measuring device-CP-MIT410-ER or the MIT420/2. The test takes 10 minutes. For this test, the test voltage must be applied to the stator for 10 minutes. After 1 minute of testing, the resistance value is recorded and stored. After 10 minutes a second insulation resistance measurement is recorded and stored. The PI value is the ratio between the 10-minute test value and the 1-minute test value.

PI value = R10 min / R1 min



The PI-measurement must be performed under the following conditions:

Stator temperature of max. 55 °C (131 °F).

Air humidity max. 75%

The winding temperature must be above the dew point during the measurement.

1. Disconnect the generator as described in section 4.1.
 - The generator has been disconnected and is de-energised.
2. Prepare the test device. Check if the battery is charged.



DANGER!



Electric current

Death or severe injury if not following proper procedure for work on the generator

⇒ Keep away from the generator during the PI test.

3. Short-circuit all phases of generator systems 1 and 2 in the rectifiers using a cable bridge.
4. Discharge the generator systems for at least 15 minutes.
5. Remove the cable bridges in the rectifiers.
6. Connect the two test leads to the measuring device.
7. Switch on the measuring device. Turn the knob to 1 kV (Fig. 9).
8. Select the PI measuring feature. Press the *Dar/PI/t* button until the letters *PI* appear on the display (Fig. 9).

5.2 PI-measurement on a generator with 2 systemsn

1. In the rectifier, measure a phase of generator system 1 against earth. To do this, keep the *Test* button pressed down for 5 seconds.
 - ➡ The measurement is performed.
2. The test finishes automatically after 10 minutes.
3. Press the *uA/v/s* button to call up the results for measurement voltage, leakage current, R1 min, R10 min and PI value (Fig. 9).
4. Discharge the generator system for 15 minutes. To do this, short-circuit all phases of generator system 1 in rectifier 1 to earth using a cable bridge.
 - ➡ The PI value for generator system 1 is measured.
5. In the rectifier, measure a phase of generator system 2 against earth. To do so, keep the *Test* button pressed for about 5 seconds.
 - ➡ The measurement is performed.
6. The test finishes automatically after 10 minutes.
7. Call up the measuring results as described in work step 3.
8. Discharge the generator system for 15 minutes. To do this, short-circuit all phases of generator system 2 in rectifier 2 to earth using a cable bridge.
 - ➡ The PI value for generator system 2 is measured.
9. In the rectifier measure a phase of generator system 1 against a phase of generator system 2.
10. The test finishes automatically after 10 minutes.
11. Call up the measuring results as described in work step 3.

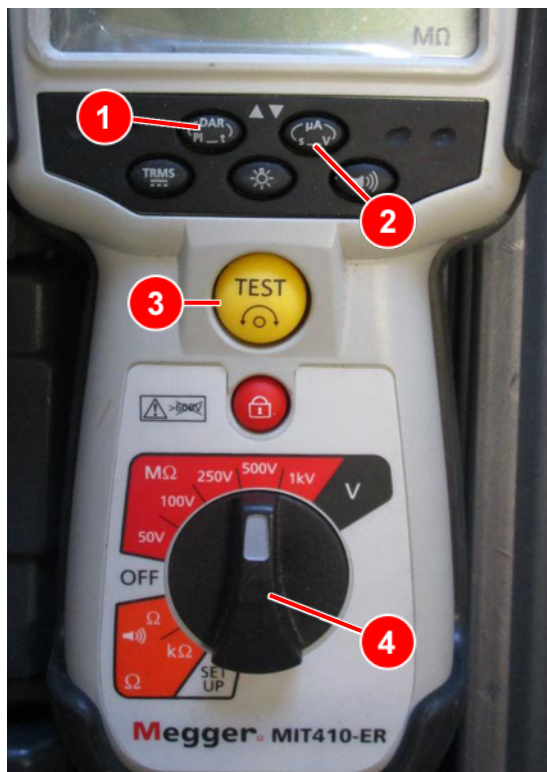


Fig. 9: PI test with insulation tester

- 1 Selecting PI test feature
- 2 Retrieving measuring results
- 3 Test button
- 4 Switching on / selection measurement voltage

12. After the PI test is complete, fill in measurement report TD-esc-03-de-en-15-009.

13. Check all measured values as per Table 6.

Measured value	Result	Work step
≥ 1	OK	-
< 1	Not OK	Create a findings report

Table 6: Results

14. Switch off the tester.

5.3 Measuring the polarisation index (PI) on an E-126 generator with 8 systems

Measuring systems 1 and 2 on generator 1 against earth

1. Short-circuit all phases of generator systems 1 to 8 in the rectifiers using a cable bridge.
2. Discharge the generator systems for at least 15 minutes.
3. Disconnect all surge arresters on the generator individually.
4. Remove the cable bridges in the rectifiers.
5. On the generator connector plates of generator 1, short-circuit a phase of generator system 1 with a phase of generator system 2 using a cable bridge.
6. Connect the test lead (red) to one of the phases of generator system 1.
7. Connect the second test lead (black) to earth (PE).
8. Switch on the measuring device. Turn the knob to 1 kV (Fig. 9).
9. Select the PI measuring feature. Press the Dar/PI/t button until the letters PI appear on the display (Fig. 9).
10. Simultaneously measure generator system 1 and system 2 against earth. To do so, keep the *Test* button pressed for about 5 seconds.
 - ➡ The measurement is performed.
11. The test finishes automatically after 10 minutes.
12. Press the *uA/v/s* button to call up the results for measurement voltage, leakage current, R1 min, R10 min and PI value (Fig. 9).
13. Disconnect the test leads, including the cable bridges.
14. Discharge generator systems 1 and 2 for at least 15 minutes by short-circuiting all of the phases of generator 1 in the rectifier with earth.



Fig. 10: Measuring system against system

Measuring system 1 against system 2

15. On the generator connector plates of generator 1, connect the red test lead to one of the phases of generator system 1.
16. On the generator connector plates of generator 1, connect the black test lead to one of the phases of generator system 2.
17. Measure generator system 1 against generator system 2 (Fig. 10).
➡ The measurement is performed.
18. The test finishes automatically after 10 minutes.
19. Call up the measuring results as described in work step 12.
20. Discharge the generator systems for at least 15 minutes by short-circuiting all of the phases in the rectifier of generator 1 with earth.
21. Repeat work steps 3 to 20 on the remaining generator units 2, 3 and 4.
22. Check all measured values as per Table 7:

Measured value	Result	Work step
≥ 1	OK	-
< 1	Not OK	Create a findings report

Table 6: Results

5.4 Measuring the polarisation index (PI) on an asymmetrical EP3 generator with 4 generator systems

1. Short-circuit all phases of generator systems 1, 2, 3 and 4 in the rectifiers using a cable bridge.
2. Discharge the generator systems for at least 15 minutes.
3. Disconnect all surge arresters on the generator individually.
4. Remove the cable bridges in the rectifiers.

Simultaneously measure system 1 and system 2 against earth

5. In the rectifier on systems 1 and 2, short-circuit a phase of generator system 1 with a phase of system 2 using a cable bridge.
6. Connect the test lead (red) to one of the phases of system 1.
7. Connect the second test lead (black) to earth (PE).
8. Switch on the measuring device. Turn the knob to 1 kV (Fig. 9).
9. Select the *PI* measuring feature. Press the *Dar/PI/t* button until the letters *PI* appear on the display (Fig. 9).
10. In the rectifier of systems 1 and 2, simultaneously measure generator system 1 and system 2 against earth. To do so, keep the *Test* button pressed for about 5 seconds.
 - ➡ The measurement is performed.
11. The test finishes automatically after 10 minutes.
12. Press the *uA/v/s* button to call up the results for measurement voltage, leakage current, R1 min, R10 min and PI value (Fig. 9).
13. Disconnect the test leads, including the cable bridges.
14. Discharge the generator systems for at least 15 min. To do this short-circuit all phases of generator systems 1 and 2 in the rectifiers to earth using a cable bridge.

Measuring system 1 against system 2

15. In the rectifier on system 1, connect the red test lead to a phase of generator system 1.
16. In the rectifier on system 2, connect the black test lead to one of the phases of generator system 2.
17. Measure generator system 1 against generator system 2. To do so, keep the *Test* button pressed for about 5 seconds.
 - ➡ The measurement is performed.

18. Press the *uA/v/s* button to call up the results for measurement voltage, leakage current, R1 min, R10 min and PI value (Fig. 9).
19. Discharge the generator systems for at least 15 minutes by short-circuiting all of the phases with earth.
20. Repeat the measurements for the remaining generator systems as per Table .

System		System/earth
3 and 4 (short-circuited with each other)	against	Earth
3		4
1 and 2 (short-circuited with each other)		3 and 4 (short-circuited with each other)

Table 8: Measurement of other generator systems

21. Check all measured values as per Table :

Measured value	Result	Work step
≥ 1	OK	-
< 1	Not OK	Create a findings report

Table 9: Results

5.5 Measuring the polarisation index (PI) on an asymmetrical E-126 EP4 generator with 8 generator systems



See sections 5.1 and 5.2 for details of how the measured values are retrieved after measurement and how to operate the device.

1. Short-circuit all phases of generator systems 1, 2, 3 and 4 in the rectifiers using a cable bridge.
2. Discharge the generator systems for at least 15 minutes.
3. Disconnect all surge arresters on the generator individually.
4. Remove the cable bridges in the rectifiers.

Simultaneously measure system 1 and system 2 against earth

5. In the rectifier on systems 1 and 2, short-circuit a phase of generator system 1 with a phase of system 2 using a cable bridge.
6. Connect the test lead (red) to one of the phases of system 1.
7. Connect the second test lead (black) to earth (PE).
8. Switch on the measuring device. Turn the knob to 1 kV (Fig. 9).
9. Select the *PI* measuring feature. Press the *Dar/PI/t* button until the letters *PI* appear on the display (Fig. 9).

10. In the rectifier of systems 1 and 2, simultaneously measure generator system 1 and system 2 against earth. To do so, keep the *Test* button pressed for about 5 seconds.
 - ➡ The measurement is performed.
11. The test finishes automatically after 10 minutes.
12. Press the *uA/v/s* button to call up the results for measurement voltage, leakage current, R1 min, R10 min and PI value (Fig. 9).
13. Disconnect the test leads, including the cable bridges.
14. Discharge the generator systems for at least 15 min. To do this short-circuit all phases of generator systems 1 and 2 in the rectifiers to earth using a cable bridge.

Measuring system 1 against system 2

15. In the rectifier on system 1, connect the red test lead to a phase of generator system 1.
16. In the rectifier on system 2, connect the black test lead to one of the phases of generator system 2.
17. Measure generator system 1 against generator system 2. To do so, keep the *Test* button pressed for about 5 seconds.
 - ➡ The measurement is performed.
18. Press the *uA/v/s* button to call up the results for measurement voltage, leakage current, R1 min, R10 min and PI value (Fig. 9).
19. Discharge the generator systems for at least 15 minutes by short-circuiting all of the phases with earth.
20. Repeat the measurements for the remaining generator systems as per Table .

System		System/earth
3 and 4 (short-circuited with each other)	against	Earth
3		4
1 and 2 (short-circuited with each other)		3 and 4 (short-circuited with each other)
5 and 6 (short-circuited with each other)		Earth
5		6
7 and 8 (short-circuited with each other)		Earth
7		8
5 and 6 (short-circuited with each other)		7 and 8 (short-circuited with each other)
1, 2, 3 and 4 (short-circuited with each other)		5, 6, 7 and 8 (short-circuited with each other)

Table 10: Measurement of other generator systems

21. Check all measured values as per Table :

Measured value	Result	Work step
≥ 1	OK	-
< 1	Not OK	Create a findings report

Table 7: Results

6 High-voltage test on generator

The high-voltage test (HV test) is used for insulation testing and diagnosing/localising faults on the generator.



An insulation test with an insulation tester must be performed prior to any HV test.

Do not perform the HV test if:

- The measured insulation resistance is 0 Ohm.
- The generator is wet due to high air humidity or rain.

6.1 Preparing HV tester



DANGER!



Electric current

Death or serious injury from electric arcs and electric shocks

- Only instructed persons may operate the HV tester and perform a high-voltage test on the generator.
- Always following the operating instructions when using the HV tester.



Fig. 11: HV tester

1. Before conducting the test, read the operating instructions of the HV tester (Fig. 11) TD-des-10-ger-eng-13-008.
2. Check the HV tester, its accessories and the cables for obvious damage and defects.
3. Connect the warning light and the foot switch (as per operating instructions).
4. Connect the test lead and the mains plug (as per operating instructions).

6.2 Conducting the high-voltage test



DANGER!



High-voltage on generator

Death or serious injury from electric current

- Keep out of the generator area during the HV test.
- Keep a safe distance from the generator during the HV test.
- Do not touch the generator during the HV test.

1. For conducting the HV test on the stator, disconnect the generator as described in section 4.1.
 - The generator has been disconnected and is de-energised.
2. Turn down the test voltage to 0 kV.



The test voltage must be set to 0 V before starting the test.

3. Set the test time on the HV tester to 60 s.
4. Set the *Prüfen/brennen* selector switch to *Prüfen* (test).
5. Set a maximum fault current of 200 mA.
6. Set the toggle switch for the measuring range to 0.2 A.
7. Insert the key-operated switch.
8. Switch on the HV tester.
 - The green indicator light is on.
9. Turn the key-operated switch to *Betrieb* (operation).
 - The red indicator light is on. The device is ready for operation.
10. Service technician 1 operates the HV tester. Service technician 2 holds the probe tip to the generator.
11. Perform the tests as per Table. 13

Defect type	Work step
Earth fault	Test phases of system 1 or system 2 against earth.
Phase-to-phase fault	Measure system 1 against system 2.

Table 8: Test types



The test starts after the foot switch has been pressed. If no fault is present, the HV tester switches off automatically after 60 s.

12. Service technician 1 presses the foot switch.



The test voltage must not exceed 1000 V.

13. Service technician 1 slowly turns up the test voltage while observing the electric current indicator.

14. If a fault is present in the generator, the current indicator needle will move unevenly. There may be a crackling sound as well as sparks and smoke. If any of this happens, abort the test immediately and locate the defect.



The HV tester stores the error automatically.

15. To reset the error, follow these steps:

- Press the foot switch.
- Turn down the test voltage to 0 V.
- Turn the key-operated switch to *Reset*.

16. If the leakage current exceeds the specified maximum fault current, the HV tester switches off automatically. It emits an audible whistling sound.

➔ The test device stores the error.

17. Reset the error on the HV tester (see work step 15).

18. End the HV test after the fault has been located.

19. Switch off the HV tester.

20. Contact Technical Service Dispatch to determine which repair work is required.

7 Finishing steps

1. Connect the surge arresters.
2. Connect the generator to the rectifiers.
3. Tighten the bolt connections to the corresponding torque according to Table 14.

Connection point	Bolt size	Tightening torque for hot-dip galvanised bolts	Torque for brass bolts
Rectifier: Copper busbar -> generator cable	M10	40 Nm	
Rectifier: Copper busbar -> generator cable	M12	61 Nm	40 Nm
Rectifier: Copper busbar -> generator cable	M16	120 Nm	90 Nm
Rectifier: Copper busbar -> Shunt resistor-copper busbar	M20	85 Nm	Not present
Generator connection plate: Copper busbar -> generator cable	M16	120 Nm	90 Nm
Generator connection plate: Copper busbar -> generator cable	M20	200 Nm	120 Nm

Table 9: Special torque for generator cable connections

4. Slide the circuit breaker panels into the filter cabinets.
5. Release the rotor lock.
6. Switch on the main switch on the nacelle control cabinet and deactivate emergency switching off on the nacelle control cabinet or nacelle sub-distribution system.
7. Start up the WEC.

8 Documenting the work performed

8.1 Service product / operation key

Two systems – the MAM system and the WEC Journal database – are currently used to process Service orders. The MAM system generates orders through SAP (known as SAP orders). These SAP orders replace the WEC Journal database.

8.2 SAP order

The activities described in this document are **not** part of regular maintenance work. A separate SAP order is required to carry out the work.

Book the following details to the Service project number of the order:

- Required materials
- Time needed

8.3 WEC Journal database

The activities described in this document are **not** part of regular maintenance work.

Record the following details in the service report:

- Required materials
- Time needed
- The **Operation key 8540**
- Document number TD-esc-01-de-xx-14-032

Upon completion, mark document number TD-esc-01-de-xx-14-032 as completed in the WEC Journal database (if applicable).

9 Points of contact

Please address any questions or issues regarding this document to the Technical Service Dispatch office in charge. If necessary, Service Dispatch can contact the ESC Electrical Engineering department.